

Muhammad Blal

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EDUCATION

Iowa State University

Bachelor of Science in Computer Science (GPA: 3.50/4.0)

Expected Graduation: **December 2027**

Ames, IA

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, JavaScript, TypeScript, SQL, HTML, CSS, Dart, Swift

Frameworks: Angular, React.js, Next.js, Node/Express.js, Flask, Spring Boot, Quarkus, FastAPI

Cloud/DevOps: Azure, AWS, EKS, Kubernetes, Terraform, Docker, Helm, CI/CD, Git, Postman, REST APIs

ML & Data/AI: Scikit-learn, Pandas, NumPy, Matplotlib, Ollama, GitHub Copilot, Claude Code, ChatGPT Codex

Databases: MongoDB, PostgreSQL, Neo4j, Elasticsearch

EXPERIENCE

Software Engineer Intern

Principal Financial Group

May 2026 – Present

Des Moines, IA

- Developed **Terraform** modules, **ArgoCD** pipelines, and **TypeScript/JavaScript** automation for **AWS EKS** infrastructure, supporting workloads for **18,000+ employees** and collaborating with the **Site Reliability Engineering (SRE)** team on production operations.
- Converted **25+ Kyverno ClusterPolicies** to **CEL** and integrated **40+ CloudWatch/Dynatrace alerts** with ServiceNow and xMatters, reducing manual EKS configuration and incident routing effort by **30%** across **5+** engineering teams.

Software Engineer Intern

J.B. Hunt Transport Services, Inc.

May 2025 – Aug 2025

Lowell, AR

- Engineered a net-new **Angular/TypeScript** feature for J.B. Hunt 360 personnel and driver panels, improving usability for **35,000+ daily users** and delivering production-ready UI functionality within a large-scale enterprise application.
- Enhanced **Java microservices** using **Spring Boot, Quarkus, MongoDB, and Elasticsearch**, with **Kubernetes CI/CD canary deployments** enabling safe releases and contributing to projected savings of **\$250,000 over 5 years**.

Software Engineering Simulation (Forge Program)

JPMorgan Chase

Jun 2025 – Jul 2025

Remote

- Designed and deployed a **Kafka-based** transaction processing system in **Spring Boot**, improving transaction validation speed by **30%** and **ensuring 100%** data integrity in an **H2 SQL database**.
- Integrated an **external Incentives API** to dynamically apply rewards, optimizing real-time balance updates and enhancing transaction accuracy.

Web Application Developer

XO Foods (Chester's)

Aug 2024 – Mar 2025

Waterloo, IA

- Built and maintained a full-stack restaurant website using **React.js, Node.js, and Express**, developing responsive customer-facing features that helped increase online orders by **35%** and improve the overall digital ordering experience.
- Optimized **responsive UI components, SEO, and Google Analytics** tracking to improve website performance and visibility, increasing mobile engagement by **50%** and monthly traffic by **20%**.

PROJECTS

Bitcoin Price Predictor | Python, Scikit-learn, Pandas, Matplotlib, yFinance

Jul 2026 – Aug 2026

- Built a machine learning model with lag features and technical indicators, improving price prediction accuracy by **15%** compared to baseline methods.
- Automated data via **yFinance API** and built iterative forecasts for future Bitcoin price trends, with outputs saved as CSV/PNG visualizations.

QuestBuddy | Java, Spring Boot, MySQL, REST APIs, WebSocket, Stripe

Aug 2025 – Dec 2025

- Built backend and full-stack integrations for a productivity platform, developing RESTful services across **10 application domains** with **Java, Spring Boot, MySQL, and JPA/Hibernate**.
- Engineered production-style features including **Spring Security/BCrypt authentication**, real-time **WebSocket** notifications, direct messaging, and **Stripe** premium subscription payments.

EasyCal | React, Node.js, Express, MongoDB, FastAPI, Python, Ollama

Feb 2025 – May 2025

- Built a full-stack nutrition and recipe planning web app used by **Iowa State University students**, featuring dynamic food search, recipe tracking, restaurant listings, and full **CRUD operations**.
- Integrated an AI-assisted recipe generator using **Ollama LLM** and **FastAPI**, boosting user engagement by over **40%** and reducing manual content entry by **50%**.

LEADERSHIP

Computer Science and Software Engineering Club (CSE) | Member

Northeast Iowa Food Bank | Volunteer

September 2024 – Present

October 2020 – Present